

# Notes 7

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## Wildcards

Asterisk \* – Matches zero or more characters

### Usage:

Use \* anywhere in a filename pattern to match any sequence of characters.

**Example 1: List all .txt files**

```
ls *.txt
```

**Example 2: Delete all backup files**

```
rm *.bak
```

**Example 3: Copy all files starting with "report"**

```
cp report* ~/backups/
```

Question Mark ? – Matches exactly one character

### Usage:

Use ? to match any single character in that exact position.

**Example 1: List files like file1, fileA, file\_**

```
ls file?
```

**Example 2: Remove all 3-character files**

```
rm ???
```

**Example 3: View contents of doc1, doc2, doc3**

```
cat doc?
```

Square Brackets [] – Matches one character from a set

### Usage:

Put allowed characters or ranges (like 0-9 or a-z) inside [] to match one character from the set.

**Example 1: List .jpg or .png files**

```
ls *.jpg
```

**Example 2: Remove test1, test2, or test3**

```
rm test[1-3]
```

**Example 3: List files starting with a vowel**

```
ls [aeiou]*
```

Negated Brackets [!] or [^] – Matches anything NOT in the set

**Usage:**

Use ! or ^ inside brackets to match any character except those listed.

**Example 1: List files that do NOT end in .txt**

```
ls *.[!t][!x][!t]
```

**Example 2: Remove files that do NOT start with a digit**

```
rm [!0-9]*
```

**Example 3: List files whose second character is not a lowercase letter**

```
ls ?[!a-z]*
```

Brace Expansion – Create entire directory structures

**Usage:**

Use {} to generate multiple strings from comma-separated lists or ranges. Combine with mkdir -p to build complex folder trees in one command.

**Example 1: Multi-folder project structure**

```
mkdir -p project/{src,docs,tests,assets}/{main,utils,config}
```

**Example 2: Versioned features with nested folders**

```
mkdir -p app/v{1..3}/feature{1..5}/{model,view,controller}/
```

**Example 3: Dated backup directory structure**

```
mkdir -p backups/{daily,weekly,monthly}/${date +%Y-%m-%d}/logs
```